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Assessing the impact of climate and land use change on flood vulnerability: a machine learning approach in coastal region of Tamil Nadu, India

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Abstract

Flooding and other natural disasters threaten human life and property worldwide. They can cause significant damage to infrastructure and disrupt economies. Tamil Nadu coast is severely prone to flooding due to land use and climate changes. This research applies geospatial tools and machine learning to improve flood susceptibility mapping across the Tamil Nadu coast in India, using projections of Land Use and Land Cover (LULC) changes under current and future climate change scenarios. To identify flooded areas, the study utilised Google Earth Engine (GEE), Sentinel-1 data, and 12 geospatial datasets from multiple sources. A random forest algorithm was used for LULC change and flood susceptibility mapping. The LULC data are classified for the years 2000, 2010, and 2020, and from the classified data, the LULC for years 2030, 2040, and 2050 are projected for the study. Four future climate scenarios (SSP 126, 245, 370, and 585) were used for the average annual precipitation from the Coupled Model Intercomparison Project 6 (CMIP6). The results showed that the random forest model performed better in classifying LULC and identifying flood-prone areas. From the results, it has been depicted that the risk of flooding will increase across all scenarios over the period of 2000–2100, with some decadal fluctuations. A significant outcome indicates that the percentage of the area transitioning to moderate and very high flood risk consistently rises across all future projections. This study presents a viable method for flood susceptibility mapping based on different climate change scenarios and yields estimates of flood risk, which can provide valuable insights for managing flood risks.

Keywords Flood, LULC, CMIP6, Random forest, Tamil Nadu

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Introduction

Natural disasters, including hurricanes, floods, droughts, and wildfires, have become more frequent and more severe globally as a result of climate change (Peresan and Hassan 2024). The increasing occurrences of natural disasters will impact human populations and the environment, with potential long-term negative impacts on communities, ecosystems, and economies. Such disasters can disproportionately affect more vulnerable communities, which should demand attention and action from governments, organisations, and individuals worldwide. The Intergovernmental Panel on Climate Change (IPCC) has noted that extreme events are becoming more frequent and intense, particularly in high latitudes and tropical regions (IPCC 2018). The United Nations Office for Disaster Risk Reduction (UNDRR) reports that these more frequent and intense disasters undermine development and endanger lives and livelihoods. From 2000 to 2019, 7348 significant documented disasters occurred globally, leading to the loss of approximately 1.23 million lives and impacting a staggering 4.2 billion individuals. Floods comprise the majority of catastrophes, 43% of all documented disasters between 2000 and 2019 (Coumou et al. 2013; UNDRR 2020). According to the World Bank, between 1980 and 2019, climate-related disasters caused \$2.3 trillion in economic losses worldwide (WHO 2021). Asia is the continent that experiences the highest frequency of natural disasters, accounting for more than 40% of all such occurrences globally (Saravanan and Abijith 2022).

In recent years, India has faced several major disasters significantly impacting the people, economy, and environment. In 2015 and 2023, heavy rains in Chennai caused floods, loss of lives, crop and infrastructure damage. Kerala witnessed its worst flooding in a century in the month of August 2018, when torrential rains created landslides and floods that resulted in over 400 fatalities and major property and infrastructure damage (Saravanan and Abijith 2022; Sundar and Kundapura 2023a). Cyclone Fani, which struck India's eastern coast in May 2019, killed over 80 people and seriously damaged infrastructure. The storm caused heavy rainfall, strong winds, and flooding in several states, including Odisha, West Bengal, and Andhra Pradesh (Mishra et al. 2021). In February 2021, flash floods and landslides caused by heavy rainfall hit the Chamoli district of Uttarakhand, leading to the death of over 50 people and causing extensive damage to infrastructure and property (Verma et al. 2022). Cyclone Tauktae in May 2021 caused heavy rainfall, strong winds, and flooding in several states, including Gujarat, Maharashtra, and Goa.

Land use and land cover (LULC) are among the major influences on the driving forces of climate change, and

they immediately affect the environment, society, and economy (Turner et al. 2007). As the rapid growth of the population results in the unsustainable exploitation of the natural resources. To evaluate the impact of human activity on the environment and create efficient policies for sustainable land management, it is essential to understand the dynamics of LULC changes. The Global Land Outlook study highlights that there has been notable environmental deterioration as a consequence of LULC change, including deforestation, soil degradation, water shortages, and biodiversity loss. There was a 3.3% decline in worldwide forest area between 1990 and 2015, while agricultural land had a 5.7% gain. The urban area has also increased by 1.3% during the same period. Due to these modifications, ecosystem services including water management, and carbon sequestration have been impacted, and greenhouse gas emissions have risen (UNCCD 2017).

India is one of the most populous countries globally, experiencing rapid urbanisation and industrialisation, resulting in significant LULC changes. According to the India State of Forest Report (ISFR 2021), the forest cover in India increased by 0.56% between 2017 and 2019. However, the report also highlights that about 24.39% of the country's geographical area is undergoing desertification and land degradation, and 17% of the forest cover is highly degraded (IPCC 2019). Marine extreme events cause significant socioeconomic damage through coastal flooding. Despite the expected increase in these events due to climate change, there needs to be more information and prediction ability when mapping coastal floods (Museru et al. 2024; Xu et al. 2024). In recent years, our societies have experienced numerous severe climatic events that have heightened public awareness of climate change's ongoing and increasing impact on daily life beyond that of prior decades. Due to the significant coastal population, the continuous increase in sea levels and its potential implications have garnered considerable interest, primarily from the scientific community, as well as stakeholders, governments, and the general public (DeConto and Pollard 2016; Vousdoukas et al. 2016; Museru et al. 2024; Jennath and Paul 2024). Multiple studies (Reynard et al. 2001; Roy et al. 2020; Pabi et al. 2021; Sugianto et al. 2022) indicate that LULC and climate change are two major contributors to the rise in flood occurrences. These changes can increase runoff and reduce natural storage capacity, making areas more susceptible to flooding. Furthermore, precipitation patterns and rainfall intensity changes can also exacerbate flood risks (Lei et al. 2021). These findings highlight the importance of considering LULC and climate change when implementing flood mitigation measures. For the susceptibility mapping studies, various machine learning models have been widely used, including GBM (Sahin

2020), XGBoost (Ma et al. 2021), RF (Vincent et al. 2023), ANN (Kalantari et al. 2019), SVM (Youssef et al. 2022), Naïve Bayes (Chen et al. 2020), Boosting Regression Tree (Mohsen Mousavi et al. 2017), CART (Basheer Ahammed and Pandey 2019), and decision trees (Mohammadi et al. 2020) but the RF model (Belgiu and Drăgu 2016; Lee et al. 2017; Abedi et al. 2021; Saleh et al. 2022; Shahiri Tabarestani et al. 2023) performed better when compared with the other models.

Climate change influences South Asia's monsoons, tropical cyclones, and floods. Temperature and rainfall changes lead to increased melting of snow and glacier ice, causing more frequent floods in the short term (Douglas 2009). Constructed environments can modify the topography, limiting the infiltration of rainwater into the soil and, therefore, amplifying surface water flow. The transition from agricultural and other land surfaces to constructed areas forms an essential part of urban development, impacting hydrology in four ways: peak flow characteristics, total runoff, water quality, and hydrologic amenities (Brenkert and Malone 2005; Rogger et al. 2017; Muttarak and Dimitrova 2019; Miller et al. 2020; Avashia and Garg 2020; Bukovsky et al. 2021a; Varughese and Purushothaman 2021; Kim et al. 2023).

The organisation of the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6) was undertaken by the World Climate Research Programme (WCRP). The CMIP6 is the latest iteration of a multi-model effort to understand better and simulate Earth's climate system. Precipitation plays a pivotal role in the climate system, and the assessment of precipitation simulations derived from CMIP6 models has significant importance in comprehending the precision and dependability of projecting future climate patterns (Sahabi-Abed et al. 2023).

Bias correction is a statistical technique widely employed in climate research to account for systematic errors or biases in both observational datasets and climate model simulations. EC-Earth3 was used to investigate the likelihood of rainfall-induced flooding during periods of intense rainfall. This model is well known for its thorough approach to climate research and high-resolution simulations. EC-Earth3 presents a comprehensive perspective on climate dynamics through integrating interactions among Earth's systems, thereby yielding a crucial understanding of the evolving climate. One notable feature of EC-Earth3 is its capacity to generate precise regional climate projections, thereby facilitating the examination of climate phenomena at more detailed spatial resolutions. The level of precision provided by this approach facilitates comprehension of localised climate variability and enables the evaluation of potential consequences of climate change on particular geographical areas. The bias correction of CMIP6 EC-Earth3 climate

model simulations is widely carried out using the Indian Meteorological Department (IMD) data for the precipitation in Indian scenarios. The most recent version of scenarios, used in the CMIP6 spanning from 2016 to 2021 and prominently highlighted in the IPCC Sixth Assessment Report (AR6) (2021), are founded upon a collection of Shared Socioeconomic Pathways (SSPs). The scenarios based on SSPs are the most intricate developed so far, including a spectrum that extends from very ambitious mitigation efforts to a continued rise in emissions. The experimental design of Scenario MIP identified four cases as Tier 1 scenarios, signifying their designation as the one with the highest priority. The Tier 1 scenarios include a wide spectrum of uncertainty in forthcoming socioeconomic and climatic forcing trajectories. They are valuable benchmarks for comparing against the RCP2.6, RCP4.5, RCP6.0, and RCP8.5 scenarios used in CMIP5. These include SSP 126, SSP 245, SSP 370 and SSP 585. The SSP 126 represents a future with rapid transition towards sustainable development and a strong emphasis on reducing greenhouse gas emissions, yielding a low-carbon economy. The SSP 245 represents a middle-of-the-road scenario, where emissions continue to rise but at a slower rate, and efforts are made to transition towards more sustainable development. The SSP 370 represents a future where emissions continue to rise at historical rates, with limited action taken to mitigate climate change, and society continues to prioritise economic growth over sustainability. The SSP 585 represents a future where emissions continue to rise even faster, with little or no action taken to mitigate climate change, and society focuses on maximising economic growth over sustainability (IPCC 2021). Reddy and Saravanan (2023) compared 17 CMIP6 datasets with data from the Indian Meteorological Department (IMD) to analyse extreme precipitation events in India. The study evaluates numerous extreme precipitation indices and concludes that the EC-Earth 3 model performs well. Therefore, the EC-Earth 3 data are incorporated into the present study.

This research aimed to create a flood susceptibility map for the Tamil Nadu coast in India, which is prone to flooding. Previous studies have used statistical methods to map flood susceptibility in the same region, but there is limited research applying machine-learning techniques. Additionally, previous studies have only focused on specific coast areas, not the entire region. This study explored the impact of LULC and climate change on flood susceptibility in the region and predicted flood susceptibility for different time periods up to 2100. The study employed 12 geo-environmental parameters, including LULC and climate scenario, as dynamic parameters. The random forest algorithm was used to identify and map flood-prone areas. The performance of the algorithm was

assessed using accuracy and receiver operating characteristic (ROC) curve metrics.

Material and methods

Study area

The study area is the coastal region of Tamil Nadu, India (Fig. 1). The region has a total area of 21,220 km² and a coastline length of 1076 km, making it India's second-longest coastline. The Tamil Nadu coastline is adjacent to the Bay of Bengal, the site of some of the world's strongest and deadliest tropical cyclones (Abijith et al. 2023). Along the coast are sandy beaches, rocky coasts, muddy flats, and the world's second-largest mangrove forest, Pichavaram, in the Cuddalore district near Chidambaram (Saranya and Saravanan 2021). The study area has a semi-arid tropical monsoon climate, with a maximum temperature of 31.5 °C and a minimum of 22.4 °C. The region receives the most rainfall between October and December (80%) after the monsoon season. Based on predictions over 130 years, precipitation is increasing during the monsoon season in the northeast and decreasing in the southwest (Bal et al. 2016). The area extending 25 km

inland from the coastline is considered in the study. This region contains major commercial and industrial hubs such as Chennai, the state capital, and 15 major ports and harbours. It is also located in the Gulf of Mannar along the Palk Strait, which serves as a maritime boundary with Sri Lanka. Despite its importance, the study area is highly vulnerable to natural disasters like tropical cyclones and tsunamis, as demonstrated by the 2004 Indian Ocean earthquake and tsunami, which killed over 10,000 people and displaced hundreds of thousands (Abijith et al. 2021).

Data analysis

A spatial database was created from multiple sources to model LULC, climatic data, and floods. As shown in Table 1, these parameters are derived from various sources. LULC maps for 2000, 2010, and 2020 were created using Landsat 5, 7, and 8 data, respectively. The Shuttle Radar Topography Mission (SRTM) DEM is downloaded from the GEE with a resolution of 30 m is used for the study. The DEM derives the Topographic Wetness Index (TWI), Stream Power Index (SPI), Slope, Aspect, and Drainage Density. The National Water

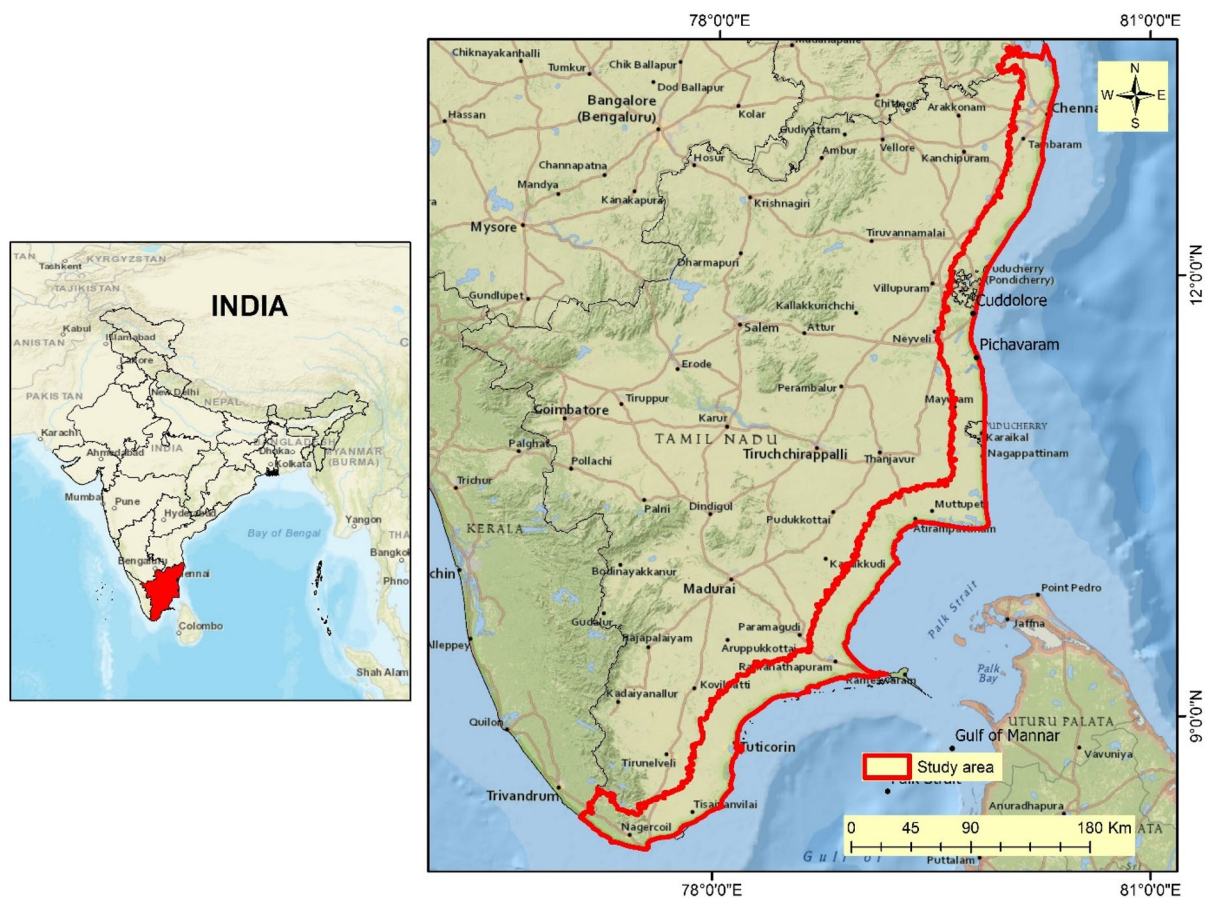


Fig. 1 Location of **a** India, **b** study area

Table 1 Data sources

Parameters	Data source	Source location
LULC	Landsat 5, 7 and 8	Google earth engine https://code.earthengine.google.com/
Height above nearest drainage (HAND)	National Water Centre	
Digital elevation model (DEM), Topographic Wetness Index (TWI), Stream Power Index (SPI), slope, aspect, and drainage density	Shuttle Radar Topography Mission (SRTM)	
Distance to coast	Landsat 8	
Rainfall	CMIP 6	https://esgf-node.llnl.gov/search/cmip6/
	IMD	https://www.imdpune.gov.in/
Road network	OpenStreetMap	https://www.openstreetmap.org
Distance to coast	Coastline	https://www.ngdc.noaa.gov
Geomorphology	Bhukosh	https://bhukosh.gsi.gov.in/Bhukosh/
Soil	National Bureau of Soil Survey and Land Use Planning (NBSS and LUP)	https://www.nbsslup.in/

Centre provides the height above the nearest drainage (HAND). The OpenStreetMap-derived Road Network is used as an input for the LULC prediction. The distance to the coast is calculated using data from NOAA's National Centers for Environmental Information (NCEI), formerly known as the National Geophysical Data Center (NGDC). Geomorphology was downloaded from the Bhukosh Geological Survey of India (GSI) website. The National Bureau of Soil Survey and Land Use Planning provides soil data (NBSS and LUP). The data have been transformed into the WGS 1984 UTM Zone 43N projected coordinate system before being fed into the machine learning model.

Preparation of the data

The Google Earth Engine (GEE) is a robust platform integrating remote sensing and GIS methodologies. It is a resilient platform that facilitates the execution of remote sensing-oriented research. It is a cloud-based environment that permits researchers to access and analyse vast quantities of satellite imagery and other geospatial data from various sources. These include Landsat and Sentinel imageries and various open-source DEM and CHIRPS rainfall data. The platform has a wide range of applications in environmental monitoring, LULC mapping, climate change research, and disaster response (Gorelick et al. 2017; Tamiminia et al. 2020). It is helpful in classifying the LULC and flood mapping since it enables users to access and analyse enormous volumes of data utilising Google's computer power (Saravanan and Abijith 2022; Sundar and Kundapura 2023b). LULC mapping using GEE has been applied in various studies, such as monitoring land cover changes in agricultural landscapes (Liu and Seto 2008; Tassi and Vizzari 2020), mapping forest cover and deforestation (Abijith and Saravanan 2022),

and mapping urban expansion (Yan et al. 2021). GEE has also been used for flood mapping, providing high-resolution data for mapping and monitoring flood extent (Kulithalai Shiyam Sundar and Deka 2021; Parthasarathy and Kundapura 2023).

Digital elevation model (DEM): Topography from DEM is crucial for determining a catchment's overland runoff and subsurface flows and identifying areas susceptible to flooding. Accurate water depth forecasts depend on the spatial resolution of DEM data (Sanyal and Lu 2004). In this research, Shuttle Radar Topography Mission (SRTM) DEM (Fig. 2a) from the United States Geological Survey (USGS), downloaded from GEE, were used. They are open-access data with a spatial resolution of 30 m. The DEM show the study area as relatively flat and low elevation, which are characteristics that make it prone to flooding. The study area exhibits a variation in elevation, with most of the region ranging from -3 m to 95 m. Almost all of the northern regions in the study area, located within a 10 km distance from the coastline, exhibit an elevation of less than 21 m above the Mean Sea Level (MSL). The southernmost region of the study shows relatively higher elevations due to the presence of the Western Ghats.

Slope: It is determined by elevation contours, governs surface water flows and influences surface runoff. In accordance with the DEM, the slope of the terrain exhibits a minimal gradient, leading to a predominantly flat topography across the majority of the study area. The area with less slope results in flooding, whereas a high slope results in quicker runoff. The slope gradient exceeds 7 degrees exclusively in the southern portion of the region, while the remaining area exhibits a slope gradient of less than 7 degrees (Fig. 2b).

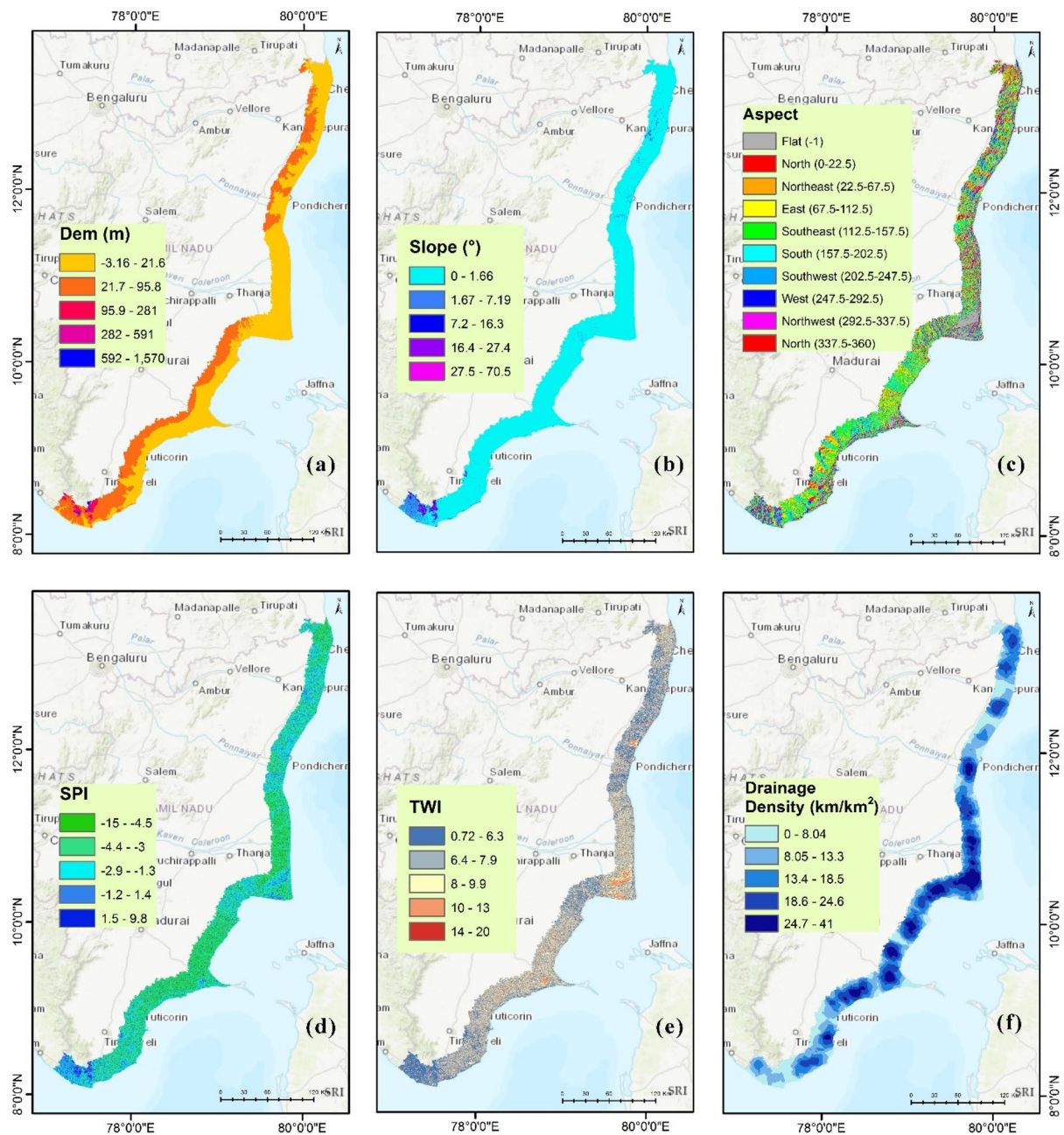


Fig. 2 Flood conditioning factors: **a** DEM, **b** slope, **c** aspect, **d** SPI, **e** TWI, **f** drainage density

Aspect: Aspect (Fig. 2c) is also essential as it affects the direction of water flow (Zzaman et al. 2021), which determines the accumulation of flooding. The aspect map indicates that the study area primarily consists of south-east, south, and east slopes, as well as flat terrain. This topographic pattern is attributed mainly to the coastal region's inclination towards the eastern part of the study area.

SPI: Stream Power Index (SPI) (Fig. 2d) measures the power of stream flow in terms of erosion and is relevant for many fluvial processes (Jebur et al. 2014). The likelihood of flooding is positively correlated with a decrease in the SPI. This phenomenon happens due to the stream's weakening force, which causes water to stagnate and, ultimately, cause flooding. A large portion of the study area exhibits SPI values that are below -1 . The southernmost

region of the study shows higher SPI values due to the presence of hilly terrain.

TWI: Topographic Wetness Index (TWI) (Fig. 2e) is a numerical measure to calculate the level of wetness in a particular area, where the wetter terrain results in more runoff resulting in flooding. The methodology relies on examining terrain topography, with a specific focus on slope and contributing area. The TWI is frequently employed in hydrological studies to forecast water's spatial distribution and potential movement within a given terrain (Das 2019; Hammami et al. 2019). The central portion of the study area, characterised as the deltaic region resulting from the River Cauvery, exhibits higher TWI values. The region located furthest to the south in the study area, characterised by its hilly topography, shows the lowest TWI value due to its steeper slope gradient.

Drainage density: The drainage density is extracted from the DEM using the hydrology tool, and the density is calculated using a spatial analysis tool in ArcGIS (Fig. 2f). Drainage density greatly influences flood severity (Swain et al. 2020). The study's river confluence region has a very high drainage density. This indicates that flooding will occur in the higher-density region due to the confluence of the drainages.

LULC: GEE provides access to Landsat 5, 7, and 8 Top of Atmosphere (TOA) reflectance images, which can be used for LULC mapping. There are several popular machine-learning algorithms available within the GEE environment. However, the random forest (RF) method has gained the most interest from the remote sensing community (Kulithalai Shiyam Sundar & Deka 2021). The LULC was classified as settlement, waterbody, barren land, forest, and vegetation using the random forest algorithm in GEE (Fig. 3).

The classification performance of the LULC maps produced from the satellite data was assessed using a classification error matrix (Abdi 2019). For 2000, 2010, and 2020, the classification error matrix was created for the validation points and the classified data. The Kappa coefficient (KC), more commonly referred to as Cohen's Kappa, is a quantitative measure employed for evaluating the precision of classification models encompassing the classification of LULC. The metric quantifies the degree of agreement between the predicted and actual classifications while also attempting to consider the possibility of chance agreements. Consequently, the KC can be used to assess and compare image classification accuracy (Kulithalai Shiyam Sundar and Deka 2021; Abijith and Saravanan 2022). The classification error matrix was also used to derive overall accuracy (OA) for each

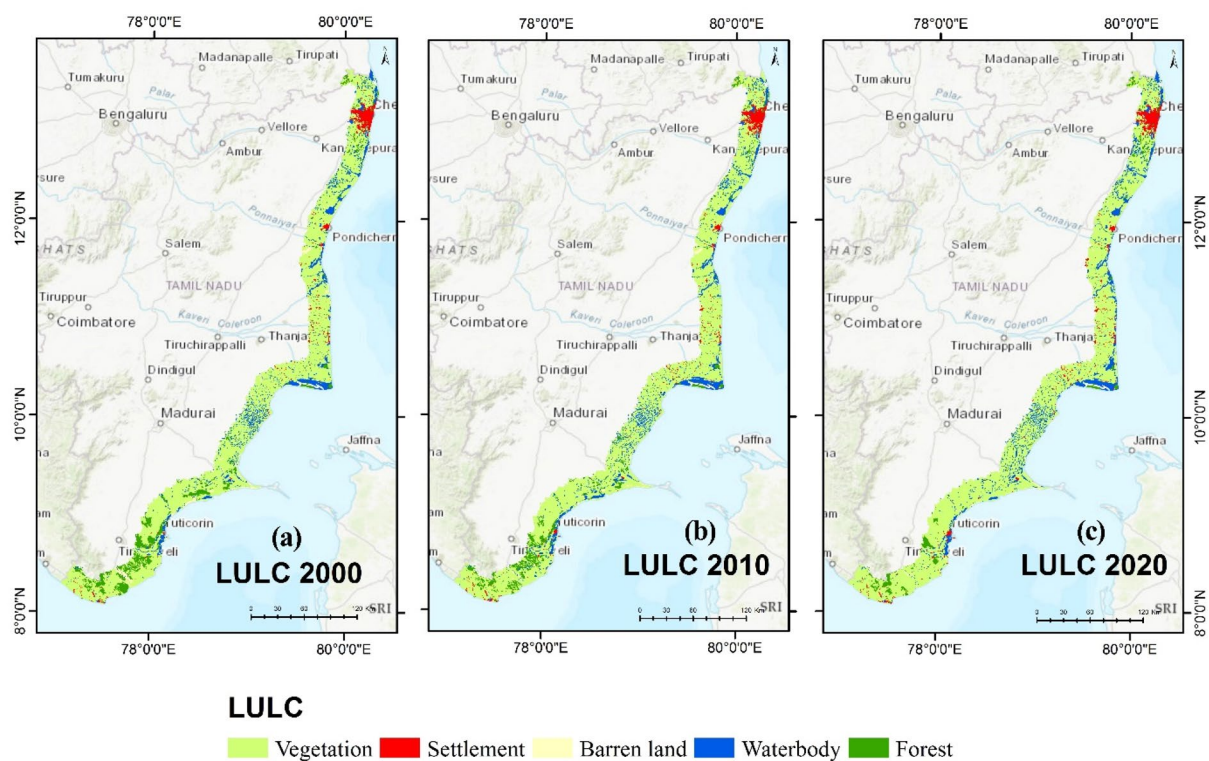


Fig. 3 LULC map of Tamil Nadu coast for the years: **a** 2000, **b** 2010, and **c** 2020

classification. The metric of Overall Accuracy quantifies the ratio of accurately classified samples, encompassing both true positives and true negatives, relative to the total number of samples present in the dataset. The provided evaluation comprehensively analyses the classification model's performance across all categories.

Soil: The soil properties (Fig. 4a), especially soil moisture, can impact flood severity. Impermeable rock reduces infiltration and enhances runoff (Saravanan and Abijith 2022). The clay soil and clayey loam soil dominate the study area, where the infiltration rate is slower, resulting in more runoff during the floods.

Geomorphology: It is essential for understanding and predicting floods. Geomorphology (Fig. 4b), the study of Earth's landforms and the processes that shape them, can shed light on the elements that determine flood occurrence, magnitude, and impact (Piacentini et al. 2020). The pediment pediplain complex is a gently sloping surface covered by a thin layer of sediments. These affect the infiltration rate of the region. The coastal plain is another geomorphological feature present in the area, which is prone to flooding because it is low-lying and close to water bodies like oceans, seas, or big rivers.

Distance to coast: Distance to the coast (Fig. 5a) plays a crucial role in determining coastal flooding (Al-Hinai and Abdalla 2020). During heavy rainfall and storm surges, the areas closest to the beaches are the most affected by flooding. The coastline is extracted from the Landsat 8 data using GEE. A Euclidean distance tool was used to evaluate a grading approach for the coast.

HAND: Height above the nearest drainage (HAND) (Fig. 5b) is valuable for flood risk assessment because knowing the vertical distance between a given location and its nearest outlet can be useful in identifying areas where water is likely to accumulate during a flood event, which may not be directly adjacent to a river or other water body. For example, a depression or low-lying area in the terrain could act as a holding basin during a flood event, accumulating water from nearby streams and resulting in flooding (Li et al. 2023).

Rainfall: The IMD provides a detailed rainfall dataset constructed from gauge readings. Spanning from 1901 to 2021, the dataset has a spatial precision of $0.25^\circ \times 0.25^\circ$. The dataset's fine resolution was realised using Shepard's interpolation method, utilising data from 6695 gauge stations (Pai et al. 2014). After 1951, data from roughly

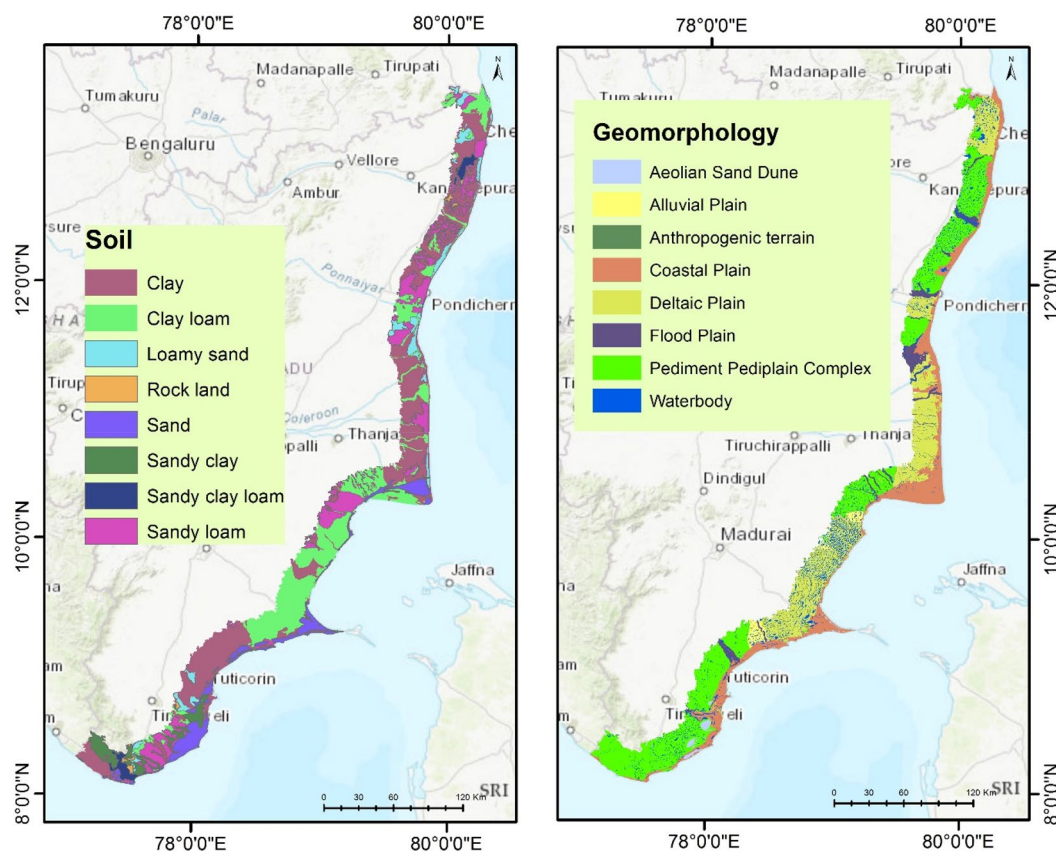


Fig. 4 Flood conditioning factors: **a** soil, **b** geomorphology

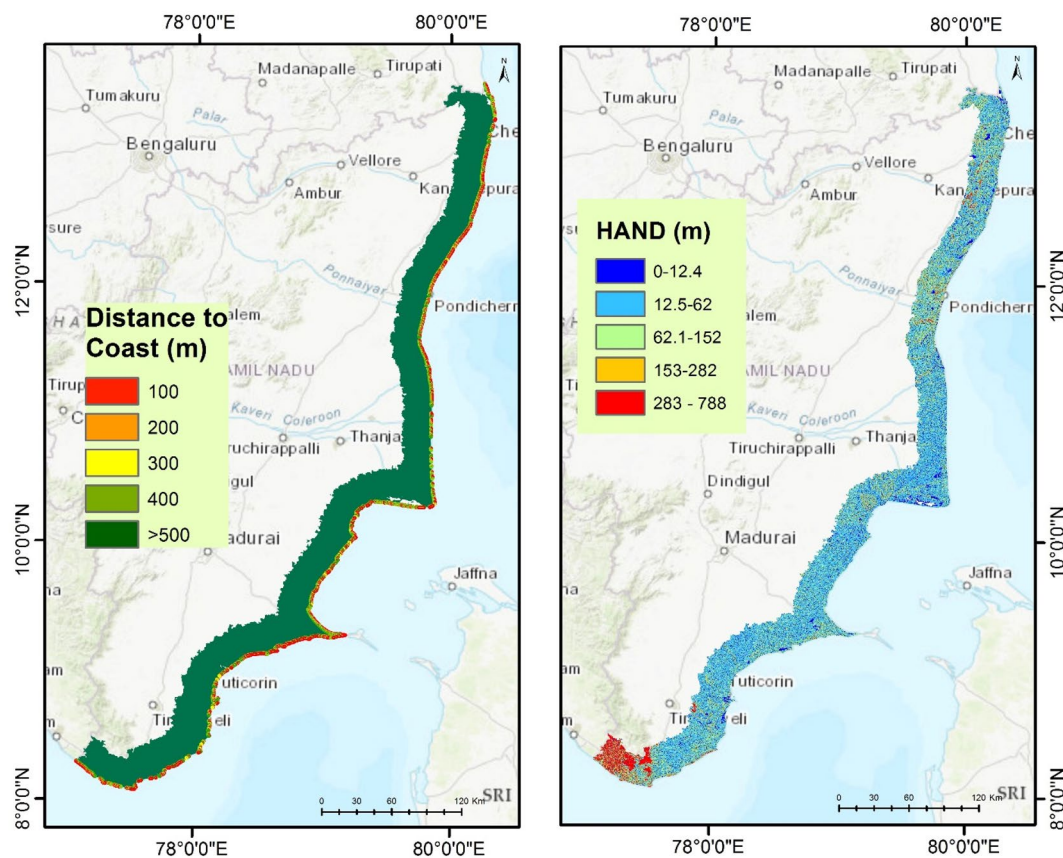


Fig. 5 Flood conditioning factors: **a** distance to coast, **b** HAND

3100 daily stations were included in the compilation. This foundational rainfall dataset (Fig. 6) for India helps counteract the inaccuracies detected in the CMIP6 models. The EC-Earth3 model is a cornerstone of climate research, providing critical forecasts for various environmental assessments. Hosted on the Earth System Grid Federation (ESGF) platform, the model stands out for its accuracy in monitoring intense meteorological events, especially in areas like India (Reddy and Saravanan 2023). Its sophisticated analysis tools are designed to integrate effortlessly with the IMD grid resolution, guaranteeing reliable and uniform outcomes for a broad spectrum of scientific research. In the streamflow projection study, we relied extensively on the insights from the EC-Earth3 model. Comprehensive data sets from this model can be accessed via the ESGF's specialised portal (<https://esgf-node.llnl.gov/search/cmip6>). This study utilises data from IMD for bias correction of CMIP6 EC-Earth3 climate model simulations. To match the IMD grid's $0.25^\circ \times 0.25^\circ$ resolution, the EC-Earth3 data underwent precise spatial adjustments using bilinear transformations. By incorporating IMD data, the study aims to improve the accuracy and reliability of the CMIP6 EC-Earth3 (Fig. 7) climate

model outputs through the identification and adjustment of any systematic discrepancies or biases that exist between the model simulations and the reference IMD dataset (Reddy and Saravanan 2023). A distribution mapping method was also used for bias correction to enhance the model's accuracy. Extreme flood events were assessed using gridded precipitation data from IMD for 2000, 2010, and 2020, along with EC-Earth3 simulations for four climate scenarios: SSP 126, SSP 245, SSP 370, and SSP 585. The maximum rainfall for the period 2050–2100 was also considered in the analysis.

The workflow of the study is shown in Fig. 7. This methodology is preferred in this study because the study area is very dynamic and rapidly developing. The conditioning factors derived from DEM and physical variables remain constant over a long period. The LULC and climate variables are predicted and given as input to the model. These maps help to predict the regions that are susceptible to floods. On the other hand, the numerical model requires inputs such as field data like runoff provided after the disasters' mishappening. Thus, the methodology used in the present study helps delineate susceptible zones beforehand of any disasters and helps take necessary action

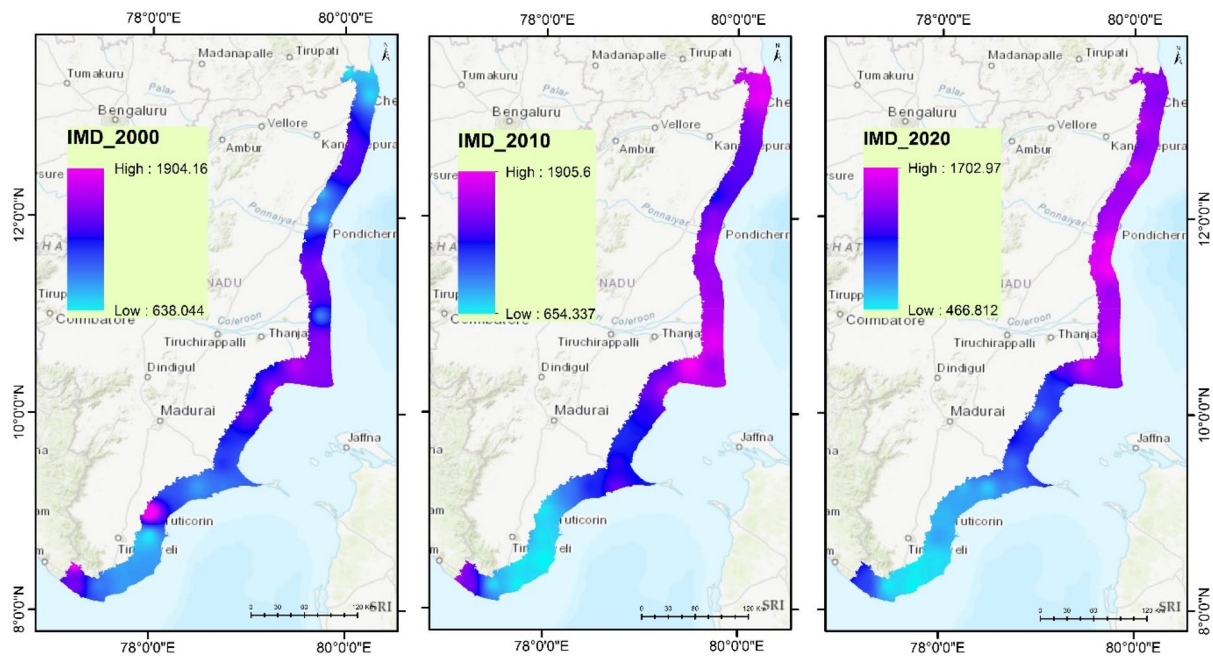


Fig. 6 IMD gridded rainfall 2000, 2010, 2020

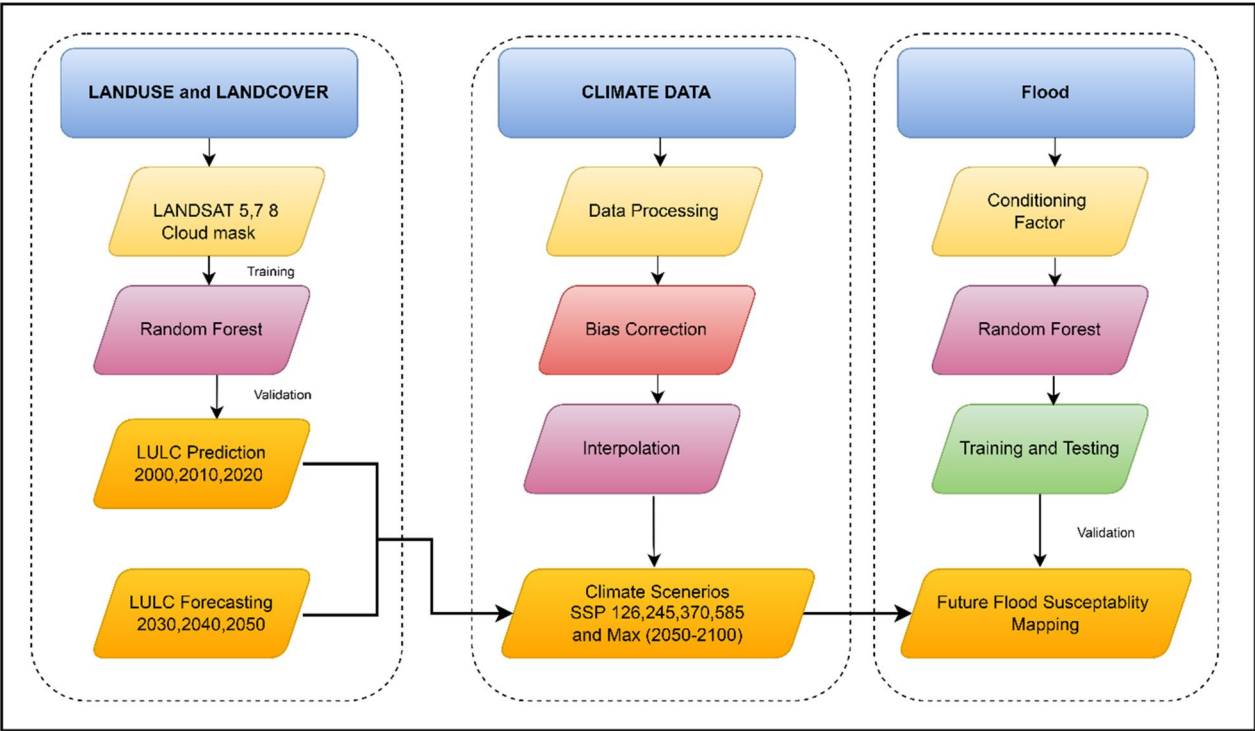


Fig. 7 Methodology carried out in the study

during the time of mishappening. This will subsequently help in saving numerous lives and properties.

LULC prediction

The Land Change Modeller (LCM) in Idrisi TerrSet software was used for LULC forecasting and identifying growth patterns and future growth scenarios for 2030, 2040, and 2050. Change analysis was performed to estimate the differences in LULC between two time intervals, i.e. 2000 and 2010. Based on the proximity theorem, Cellular Automation (CA) was used for this analysis. According to the proximity theorem, the likelihood of a pixel undergoing a transition to a specific land cover class is affected by the proximity or distance to adjacent pixels belonging to the identical class, as determined by the Markov transition rules and their surrounding neighbourhoods (Shade and Kremer 2019). The cellular automation method employs Markov Chain (MC) matrices, and changes in suitability maps are created by training a multi-layer perceptron (MLP) (Mirkatouli et al. 2015; Alam et al. 2020). The MLP was employed after specifying transition classes. The data were split 70/30 for training/validation. It was necessary to investigate LULC shifts in each study area separately because there is no universally accepted set of driving factors. Only those transitions significantly impacting LULC changes were considered (Mishra and Rai 2016). Distance from the road, 2020 settlement density, digital elevation model,

slope, the spatial trend of change from barren land to settlement, and vegetated land to settlement were the six constraint transition driver variables examined.

Flood susceptibility mapping

Feature importance using random forest scores (Fig. 8) is commonly used to identify each feature's contribution to the model's accuracy. A feature importance analysis here revealed a dominant influence by the DEM data, having the highest importance score (>0.22). This suggests a strong relationship between the DEM and the target variable.

A flood inventory map was created by identifying 4500 flood-affected pixel locations using historical flood data, field surveys, and SAR imagery. The data collected from the SAR imageries are processed using the GEE. To determine the backscatter coefficient, the Sentinel-1 SAR imagery was processed in the GEE platform, where noise was removed, radiometric calibration was performed, and the terrain was corrected. In an attempt to reduce the effect of speckle noise, the difference between two Sentinel-1 images taken before and after the flood was calculated and smoothened. The image ratio technique has been utilised to separate the flooded and non-flooded regions (Saravanan and Abijith 2022). The areas at risk of flooding were then identified using a threshold, and points were created from the frequently flooded areas. Points that were not affected by the flood were

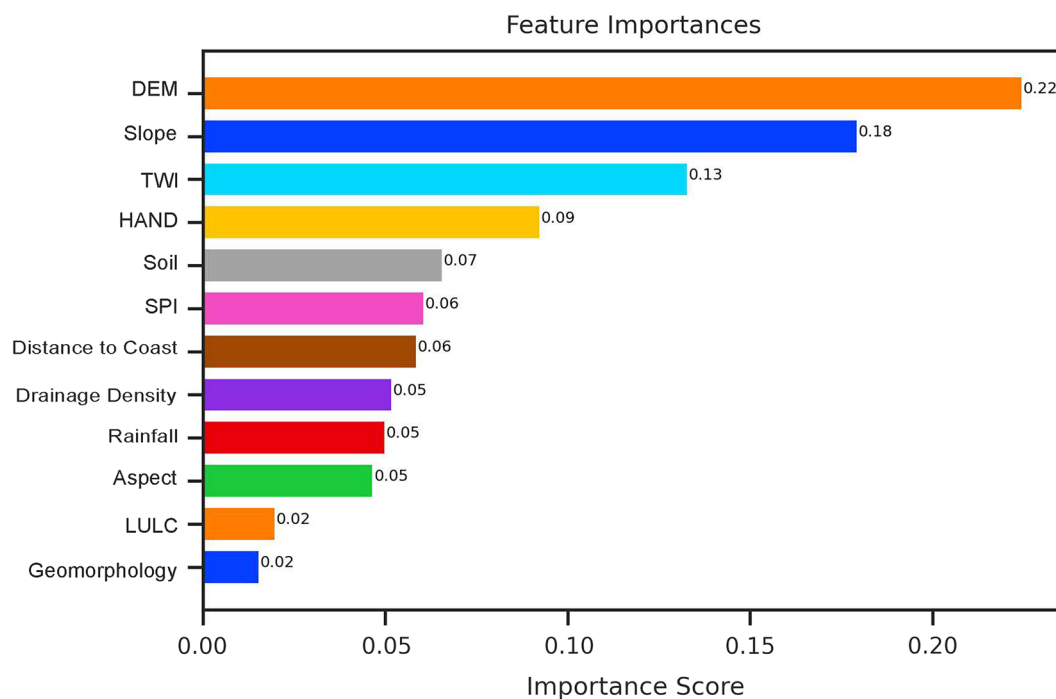


Fig. 8 Feature importance

identified using geolocation data that were obtained. Further, the flood data collected from the disaster management authority records, field visits and previous reports are also integrated and updated for the flooded inventory data preparations.

A flood susceptibility analysis was performed to identify the variables contributing to flooding, which is necessary for effective flood risk management. Here, a random forest was used to determine the variable importance. Random forest models build multiple decision trees using randomly selected subsets of the data. The algorithm then combines the results of the decision trees to make predictions. Variable importance is determined with a Gini importance score, which measures the mean decrease in model impurity that results from the inclusion of a given variable. Computing variable importance enabled the identification of critical variables that contribute the most to flood susceptibility.

Evaluation metrics

The ROC-AUC curve is a graphical representation of classification performance that was used to evaluate the flood models. It shows the trade-off between sensitivity and specificity. Sensitivity refers to the probability of correctly classifying a flooded cell, while specificity refers to the probability of correctly classifying a non-flooded cell. The area under the ROC curve (AUC) is applied quantitatively to evaluate the effectiveness of the model, with higher values indicating better performance. The false-negative rate is equivalent to the sensitivity of a flooded cell, while the false-positive rate is equivalent to the specificity of a non-flooded cell. The ROC-AUC is a statistical metric used to evaluate flood models' performance, with higher sensitivity, specificity, positive predictive value (PP), and negative predictive value (NP), indicating better model performance. These indices are commonly used to evaluate machine learning models' performance (Pham and Prakash 2017) and the equations are as follows:

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad (1)$$

$$\text{Specificity} = \frac{TN}{FP + TN}, \quad (2)$$

where TP=a count of true positive, TN=a count of true negative, FP=a count of false positive, and FN=a count of false negative.

The natural break classification method is used for the model to classify their predictions as very low, low, moderate, high, or very high. This classification method, also known as Jenks optimisation, determines the optimal value impact of various classes. It minimises the

differences within each category or group while maximising the differences between categories or groups. Areas with higher classification ratings are more susceptible to floods, whereas those with lower ratings are less likely to be affected.

Results

Land use and land cover (LULC) prediction and forecasting

LULC statistics and transition patterns helped analyse the changes and the forces behind shifts in LULC. Using the RF algorithm, the land-use transition study focused on the coast of Tamil Nadu for the years 2000, 2010, and 2020. In Tamil Nadu, water bodies include lake systems, rivers, ponds, lagoons and backwaters. The settlement areas comprise residential and industrial areas, roads, railways, and pavements. Vegetation includes plantations, cropland, and fallow land. Barren lands are sand deposits on river banks, open or vacant land, beaches, and dry lakes or ponds. Mangroves and reserve forests are classified as forests. The model's overall accuracy (OA) for 2000, 2010 and 2020 are 90.05%, 91.88% and 94.83%, respectively. The KC values are 87.32%, 89.64% and 93.06%, respectively.

An MLP neural network was used to model land cover transition potentials, representing the likelihood of future land transitioning. Driving factors were incorporated into the model with complete reporting on their explanatory power. Cramer's V correlation was used to determine the correlation between the driver variables and land use for a multi-criteria evaluation (MCE). Distance from the road had the highest Cramer's V correlation of 0.67 and was therefore identified as a critical variable that reflects the essential suitability for the transition LULC change (Table 2). The model simulated a LULC map for 2020 using the temporally adjacent image inputs. An LULC map was generated for 2020 using the driver variables and land cover images from 2000 and 2010. To test the accuracy of the projected 2020 LULC map, it was compared to the actual 2020 LULC map. The MLP predicted LULC for 2020 was compared to the classified LULC for 2020 and showed better performance of KC values of 85.6%. Thus, the model is further forecasted using the MLP to predict LULC for 2030, 2040, and 2050.

Table 2 Cramer's V for the driver variables

Driver variables	Cramer's V
Distance from road	0.67
DEM	0.56
Slope	0.34
Distance from the built-up area	0.41

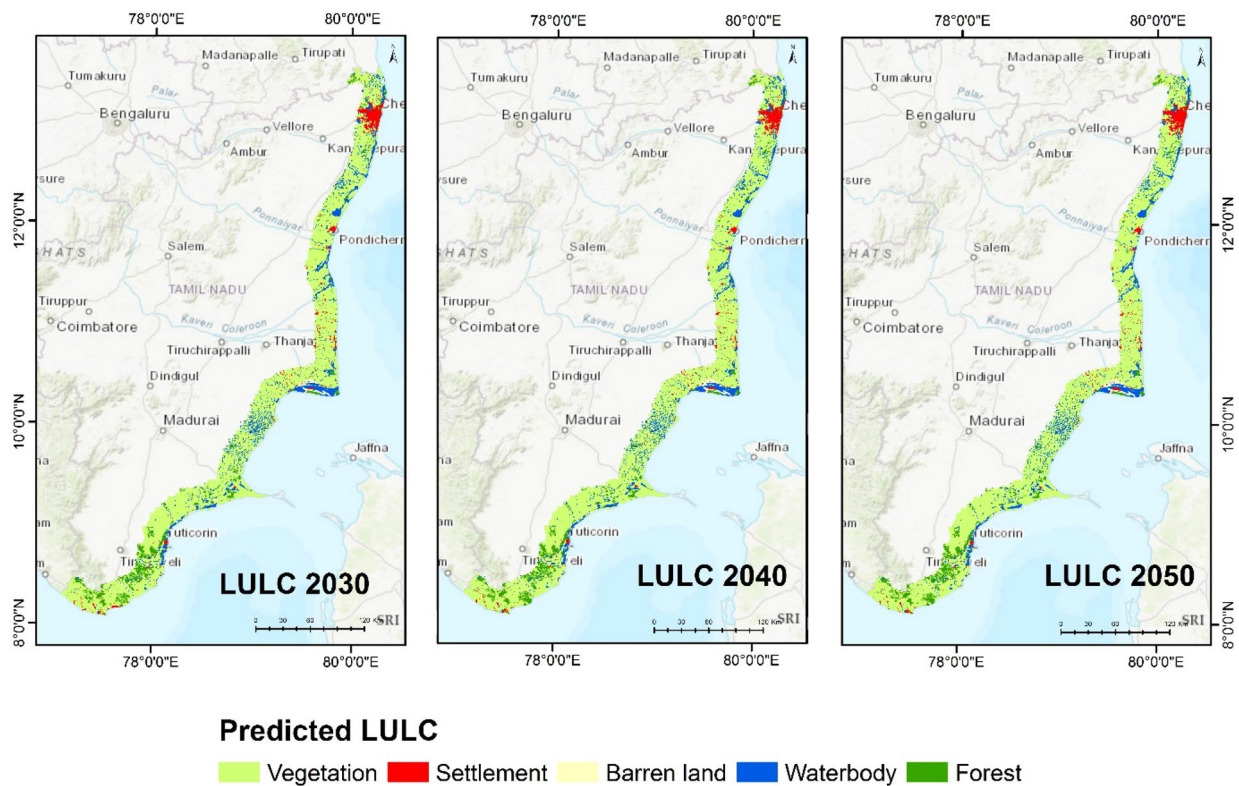


Fig. 9 Future predicted LULC of 2030, 2040 and 2050

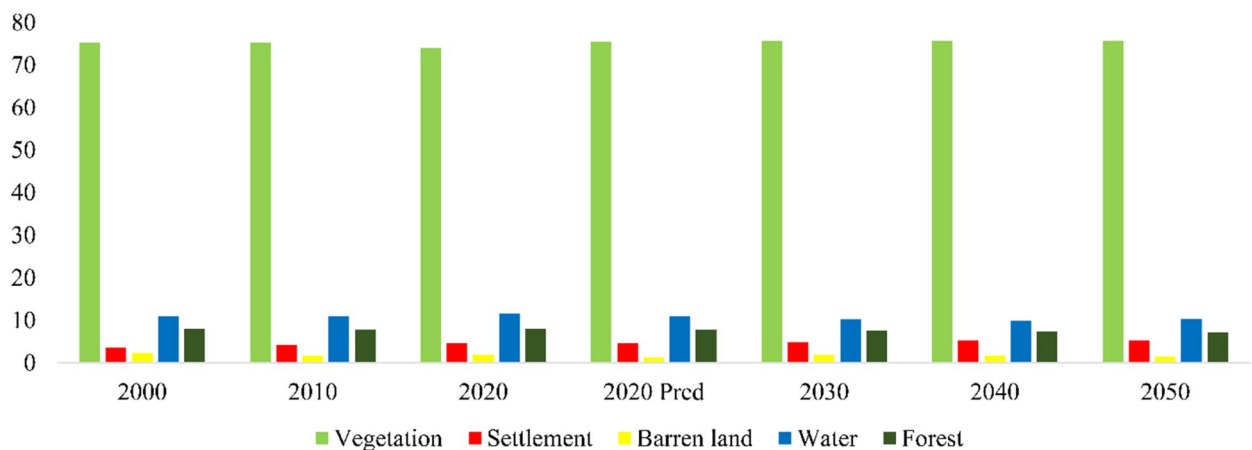


Fig. 10 Change in area of LULC

Figure 9 shows the predicted LULC for 2030, 2040, and 2050.

Figure 10 shows the changes in LULC for each class over time. From 2000 to 2050, significant changes have been seen and are expected to occur in the LULC classifications. The percentage of vegetation cover had a slight rise from 75.32% in 2000 to 75.36% in 2010, followed by

a decline to 74% by 2020. Nevertheless, projections show an anticipated rebound in 2020, with a projected increase of 75.56%. Furthermore, a steady and continuous increasing trend will persist until 2050, reaching 75.8%. The proportion of settlement areas experienced an increase from 3.49% in 2000 to 4.62% in 2020. The projected timeline from 2030 to 2050 shows a combination of changes,

ultimately stabilising at a rate of 1.46% by 2050. The proportion of water bodies showed a relatively consistent trend, maintaining a stability of 10.93% in 2000 and slightly increasing to 11.63% in 2020. However, the projected data for 2020 indicate a tiny decline to 10.91%. The anticipated estimations of the forest for 2020 indicate a potential fall to 7.73%, followed by a continuous gradual decline that is expected to reach 7.13% by the year 2050. The aforementioned trends highlight the growth of human settlements, which may come at the expense of deforestation and the degradation of barren lands. The durability of natural water reserves may contribute to the relative stability seen in water bodies. However, the reduction in forested areas raises worries over the expansion of the urban regions and the associated human-induced consequences.

Projection of CMIP6 precipitation

Figure 11 shows the bias-corrected EC-Earth3 climate model projections for 2020, 2030, 2040, 2050, and the Max(2050–2100) under four distinct scenarios.

Table 3 represents the average annual precipitation (AAP) in mm across different SSP scenarios calculated for flood prediction (Fig. 12). The SSP 126 scenario starts with an average annual rainfall of 730.12 in 2020. There is a significant increase in rainfall by 2030, reaching 1954.69. By 2040, rainfall rises to 2679.53 but then drops considerably to 1168.84 in 2050. The Max(2050–2100) shows that the average annual rainfall is projected to be 3600.13. The average annual rainfall for the SSP 245 scenario in 2020 is 1517.54. It will increase slightly by 2030 to 1743.36. However, there is a considerable decline in 2040 to 870.82, followed by a minor increase in 2050 to 1092.64. The average rainfall for the Max(2050–2100) is projected to be 3279.91. In the SSP 370 scenario, 2020 starts with a rainfall of 1868.41. A notable decrease will be observed by 2030, when the rainfall will reduce to 904.76. The rainfall then rises marginally in 2040 to 1086.03, and there's a significant increase in 2050, reaching 1975.60. For the Max(2050–2100), the average annual rainfall is anticipated to be 4407.56. The 2020 rainfall for the SSP 585 is 2311.23. It slightly decreases by 2030 to 2218.33. A steep decline is observed in 2040, where the rainfall drops to 820.49. By 2050, there's a slight increase to 1068.47. In the Max(2050–2100) scenario, the average annual rainfall is observed to be the greatest among all scenarios, with a value of 5690.62.

Random forest hyperparameter optimisation and validation

This study utilised the random forest models, which can be optimised by tuning hyperparameters, such as the number of trees, the maximum depth, and the number

of features considered at each split. Other techniques for improving performance include selecting the best optimisation method for the model. Pre-processing techniques such as normalisation or scaling of input data can also improve model performance. There is a trade-off between model complexity and performance when optimising these models (Breiman 2001). In this study, we tested three standard optimisation techniques, including (1) Grid Search (GS), which involves exhaustively searching for a predefined set of hyperparameters to identify the combination that optimises performance, (2) Random Search (RS), which involves randomly sampling and evaluating sets of hyperparameters from a predefined distribution, and (3) Bayesian Optimization (BO), which constructs a probabilistic model of an objective function and iteratively selects the hyperparameters most likely to improve performance (Bergstra et al. 2012; Snoek et al. 2012; Fernández-Delgado et al. 2014; Pk et al. 2021). These three methods were tested for mapping flood susceptibility using the classified 2020 LULC and 2020 IMD annual rainfall data. The results (Fig. 13) showed that BO (86%) outperformed GS (85%) and RS (83%). Hence, the BO algorithm is used further to model the flood susceptibility for future climate and LULC scenarios.

Evaluation of climate projections vs. flood susceptibility mapping

Flood vulnerability maps were produced for the coast of Tamil Nadu using a random forest algorithm. The maps were generated with 12 different hydrometeorological factors. The importance of these factors was determined through feature selection techniques with the random forest model. Training and validation datasets were collected from previously occurred flood scenarios to evaluate the flood susceptibility models. The training and validation datasets generated using historical flood events (2015, 2018, and 2020) and historical flood records from Disaster Management Authority reports were used as flood inventories to evaluate flood susceptibility models using the proposed technique. Flood inventory points were labelled with "1" for regions where flooding had occurred and a "0" where flooding had not occurred. A total of 4500 flood and 4500 non-flood points were generated and then divided randomly at a ratio of 70:30 for training and testing. The categorisation technique applied to the output of the ML model (described in Sect. "LULC prediction") decreased the average difference between classes while increasing the gap within each class.

SSP 126

Estimated percentages of flooded areas are shown in Fig. 14, and flood susceptibility mapping (FSM) for the SSP 126 scenario is shown in Fig. 15. Based on the

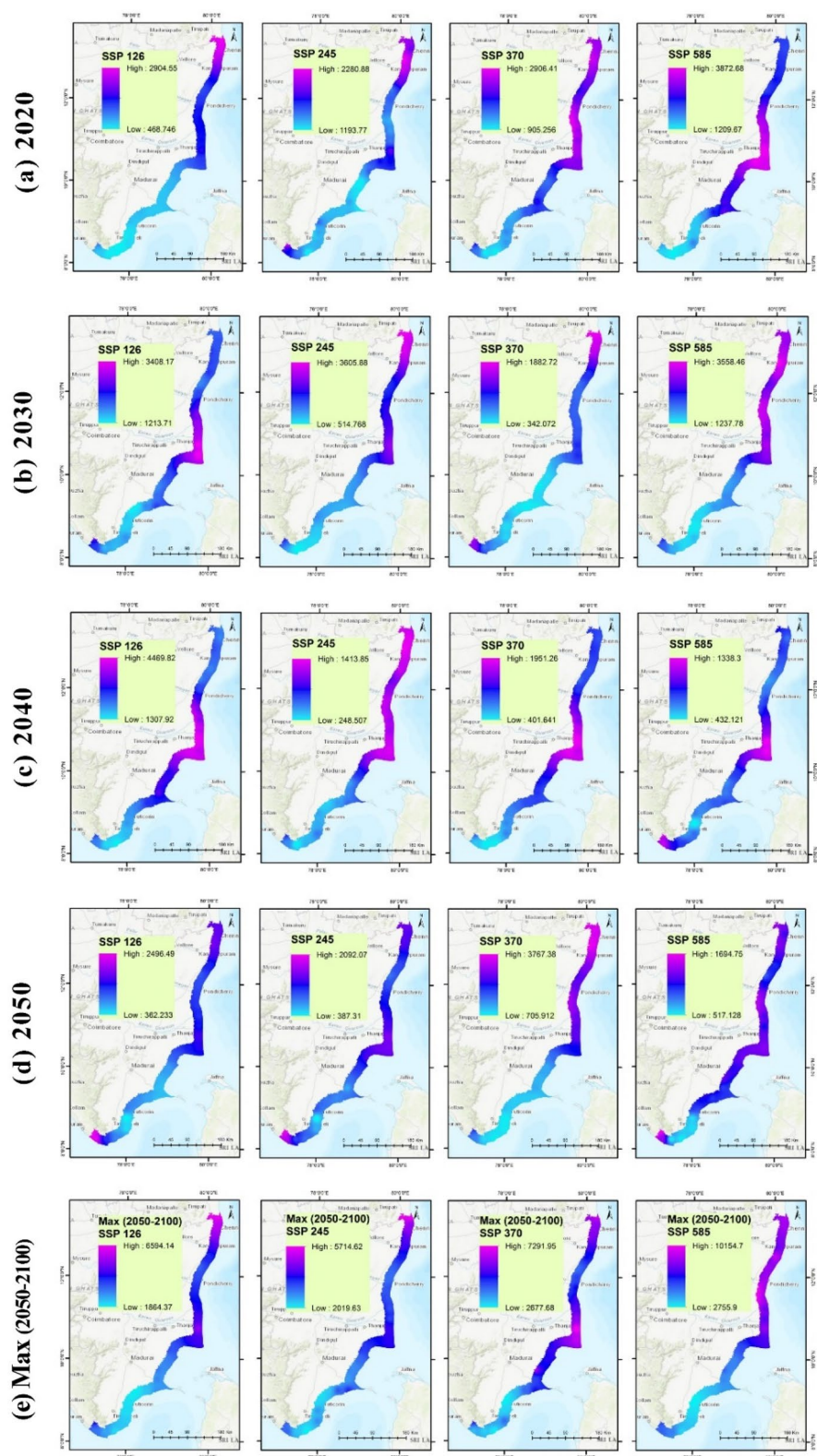


Fig. 11 EC-Earth3 projected data for years: **a** 2020, **b** 2030, **c** 2040, **d** 2050 and **e** Max(2050–2100)

Table 3 Average annual precipitation (mm) by years for CMIP6 scenarios

SSP	2020	2030	2040	2050	2050–2100
126	730.12	1954.69	2679.53	1168.84	3600.13
245	1517.54	1743.36	870.82	1092.64	3279.91
370	1868.41	904.76	1086.03	1975.60	4407.56
585	2311.23	2218.33	820.49	1068.47	5690.62

observed pattern, it can be noted that there is a decline in the percentage of areas categorised as having very low, low, and moderate vulnerability over time. Conversely, there is an upward trajectory in the percentage of regions labelled as exhibiting a very high risk in the future. The proportion of regions categorised as having moderate vulnerability experiences a decline from 27.90% in 2020 to 17.77% in 2040. Furthermore, this percentage

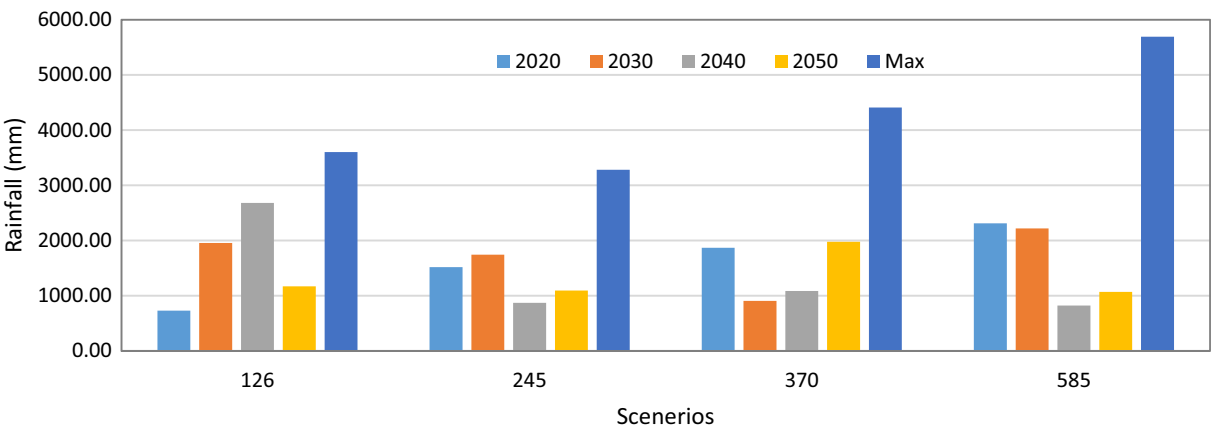


Fig. 12 Annual rainfall—SSP scenarios

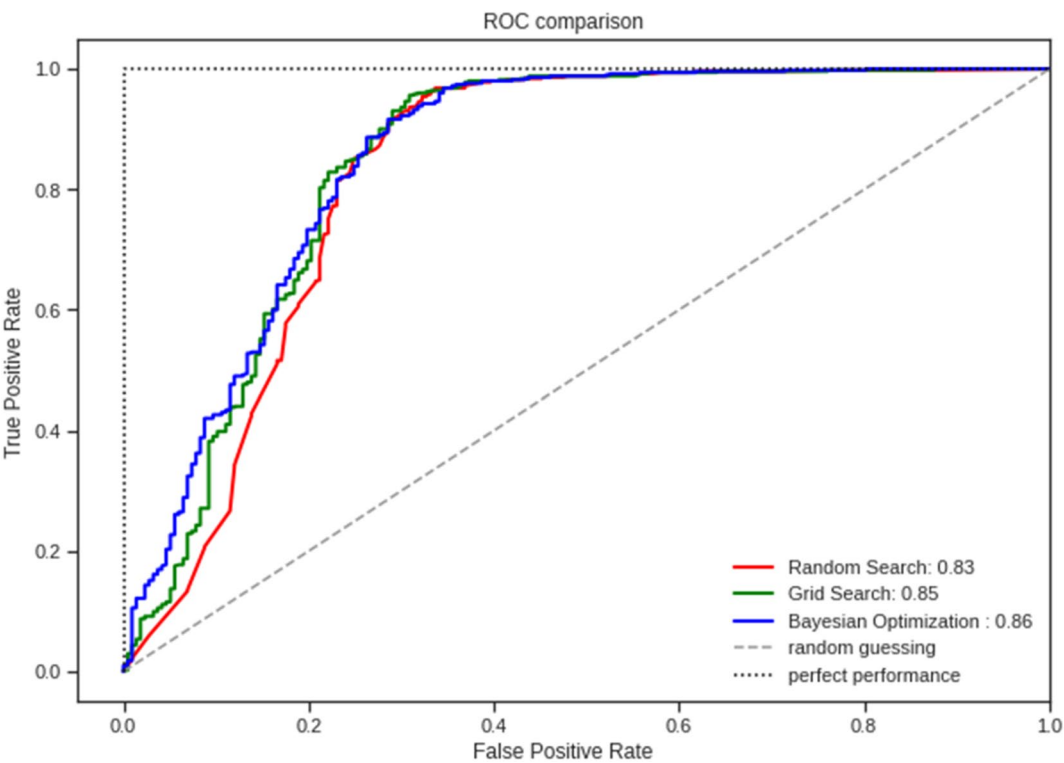


Fig. 13 Validation of random forest

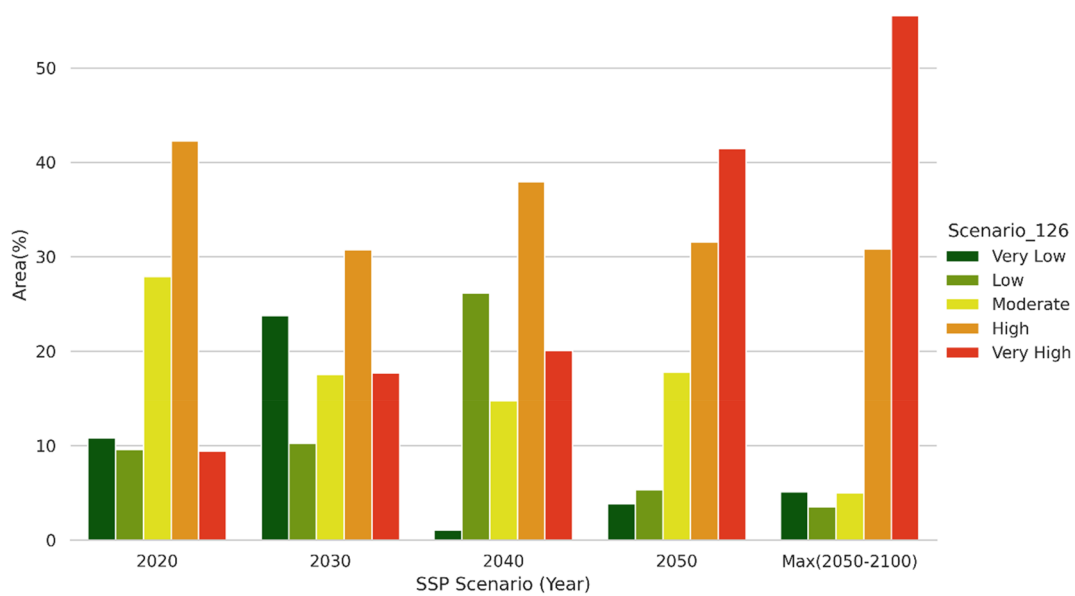


Fig. 14 Areal change of FSM for SSP-126

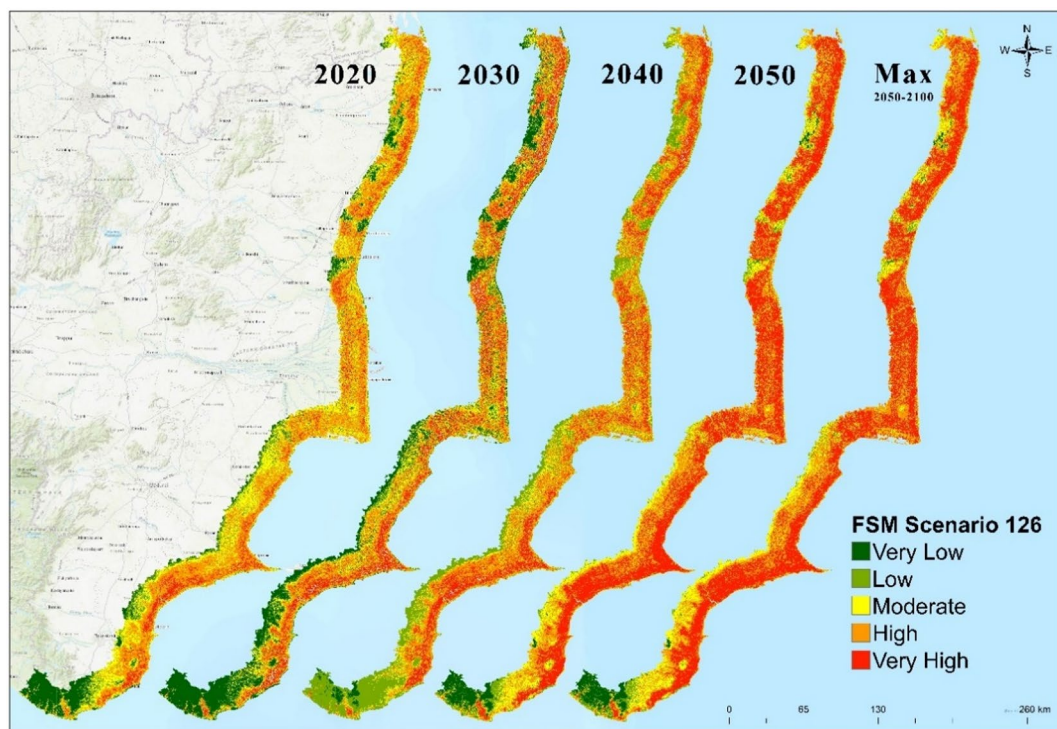


Fig. 15 Flood susceptibility map SSP-126

continues to reduce to 5.12% throughout the Max(2050–2100). The proportion of regions categorised as high-risk declines from 42.23% in 2020 to 31.56% in 2050 and has a slight decrease for the Max(2050–2100) scenario, reaching 30.78%. Overall, these trends indicate that the risk of

flooding will increase over time, with a more significant percentage of areas transitioning to moderate to very high vulnerability categories, while the proportion of areas with lower flood vulnerability will decrease. This tendency is also evident in Fig. 14, as the area classified

as having a very low vulnerability is diminishing, resulting in an increase in regions categorised as having a very high risk. The situated areas at greater distances from the coastline also see significant floods due to changes in LULC and alterations in rainfall patterns.

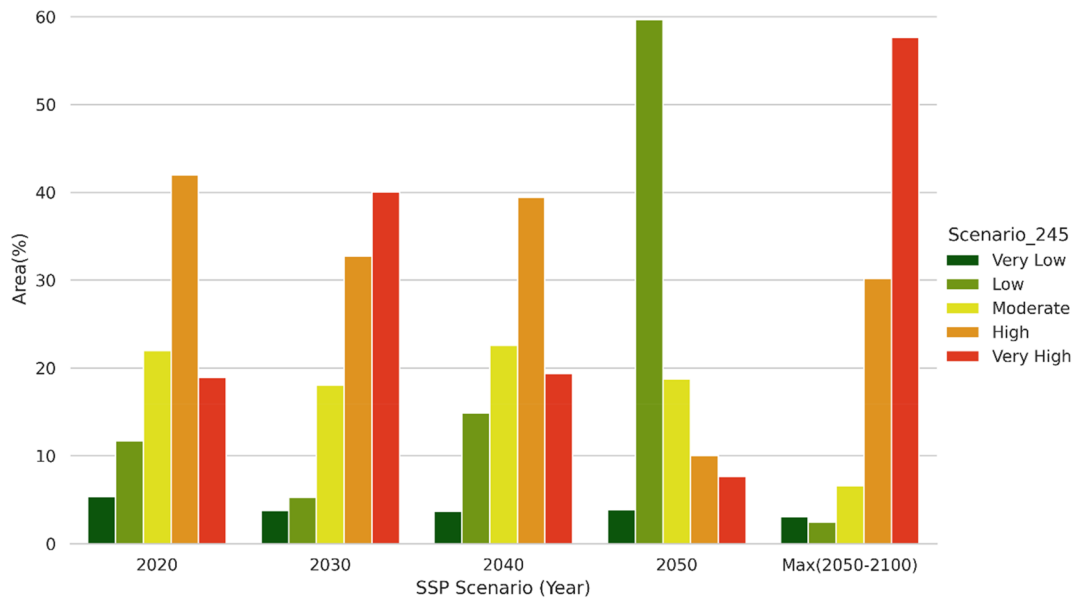


Fig. 16 Areal change of FSM for SSP-245

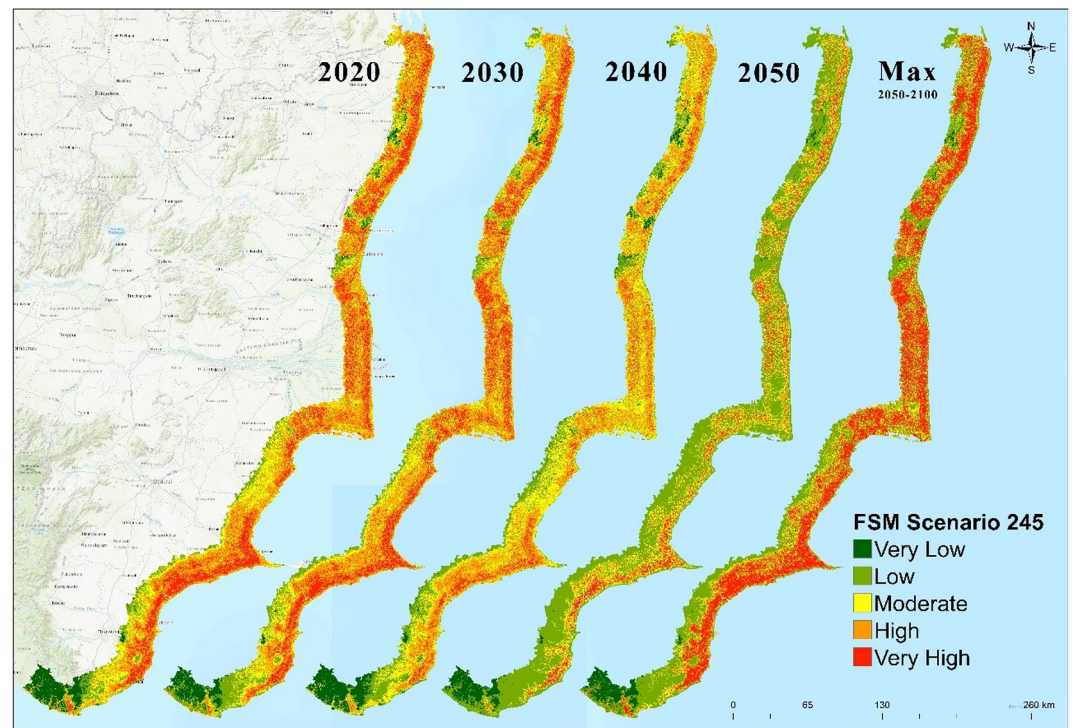


Fig. 17 Flood susceptibility map SSP-245

SSP 245

The proportions of areas impacted by flooding and FSM for the SSP 245 scenario are shown in Figs. 16 and 17, respectively. According to the data provided for SSP 245, there is a projected rise in precipitation until the year 2030. The precipitation levels show a decline between the years 2040 and 2050. During the Max(2050–2100), the highest average rainfall is seen, resulting in an increase in precipitation. Figure 16 shows increased high and very high susceptible regions to flood at 30.17% and 57.66%, respectively. Figure 17 illustrates that the inland region is also classified as a highly vulnerable area for 2020 and 2030. In contrast, the aforementioned impact is shown to undergo a decline across the time period spanning from 2040 to 2050. The conclusive analysis Max(2050–2100) reveals that the coastal districts of Tamil Nadu are situated within a region characterised by a very high level of vulnerability to flood, amounting to 57.7% of the total area. The southernmost sections of the coast comprising Tirunelveli and Kanyakumari districts are classified as areas with very low and low vulnerability in all the years for the SSP 245 scenario. The higher elevation and slopes of the Western Ghats are responsible for this phenomenon.

SSP 370

Figure 18 illustrates the distribution of regions in the SSP 370 scenario in 2020. The Max(2050–2100) analysis reveals that under the SSP 370 scenario, the annual average rainfall is recorded at 4407.56 mm. This significant precipitation level poses a potential threat, particularly

in places with very high vulnerability (58.69%) and high vulnerability (31.46%). The maps strongly correlate with the precipitation pattern of SSP 370 and LULC. The previously stated pattern is also shown in Fig. 19, whereby the FSM for the year 2030 exhibits a more significant proportion of regions categorised as having very low and low vulnerability in inland areas. However, it is noteworthy that very low vulnerability prevalence diminishes by 2040. Consequently, there was a continuing decline, leading to an increase in the area characterised by very high vulnerability for the year 2050. In conclusion, the maximum vulnerability levels observed within the study area for Max(2050–2100) indicate that a significant majority, over 90% of the region, is distinguished by very high and high vulnerability.

SSP-585

The proportions of areas impacted by floods are shown in Figs. 20 and 21 for the SSP 585 scenario. It is anticipated that the proportion of areas classified as very low, low and moderate vulnerability to floods will decline in the year 2030. In contrast, the proportion of areas classified as high vulnerability to floods will increase compared to 2020. The regions categorised as very low, low, and moderate susceptibility are projected to increase by 2040. In contrast, a decline in conditions relative to those seen in 2030 is anticipated in areas with high and very high vulnerability. The level of vulnerability, both high and very high, shows a rise in the year 2050 compared to 2040. The analysis conducted for the period spanning from 2050 to 2100, referred to as the Max(2050–2100),

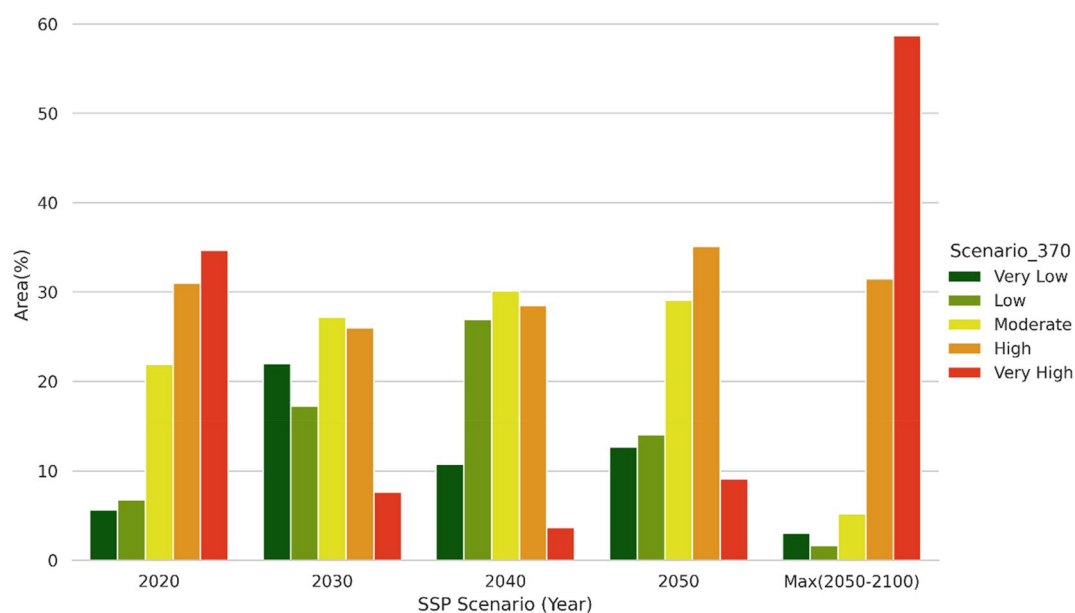


Fig. 18 Areal change of FSM for SSP-370

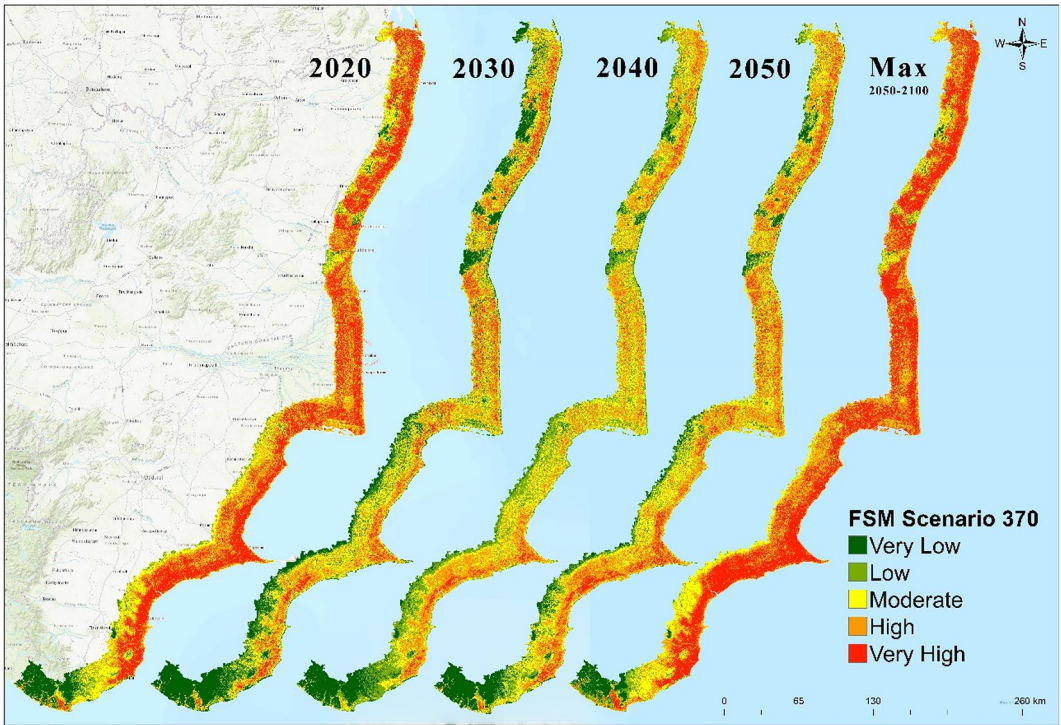


Fig. 19 Flood susceptibility map SSP-370

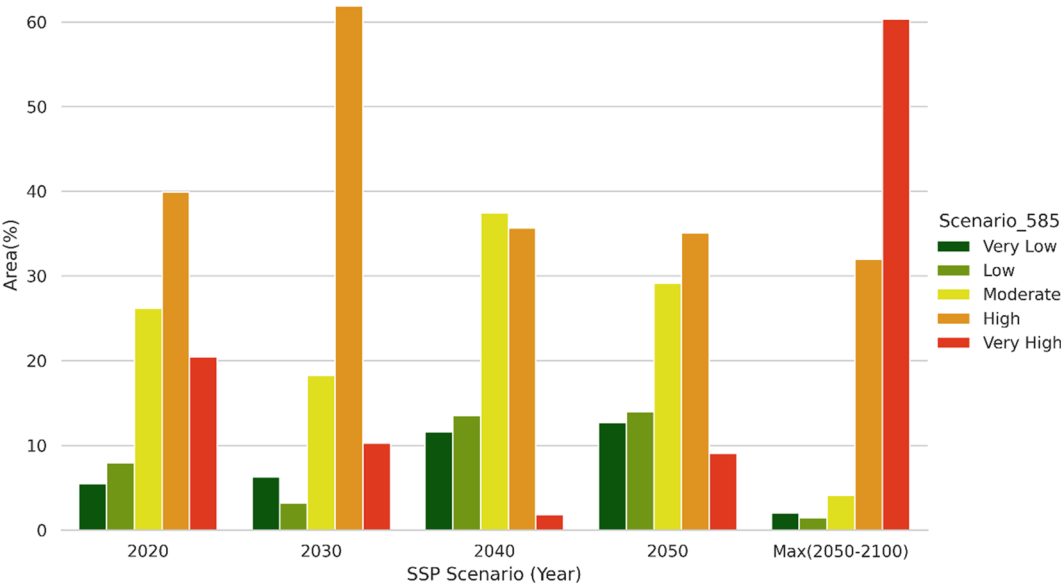


Fig. 20 Areal change of FSM for SSP-585

reveals a significant increase in the levels of vulnerability seen in places categorised as high and very high vulnerability compared to prior years. These trends also indicate that it is in sync with the fluctuating precipitation levels predicted by the SSP 585 scenario. Hence, it can be shown from Fig. 21 that there is a general downward trend in vulnerability levels until 2040, followed by an increase by 2050. The study area exhibits significant levels of vulnerability for the years 2020 and 2030, whereas for the years 2040 and 2050, vulnerability is seen to be

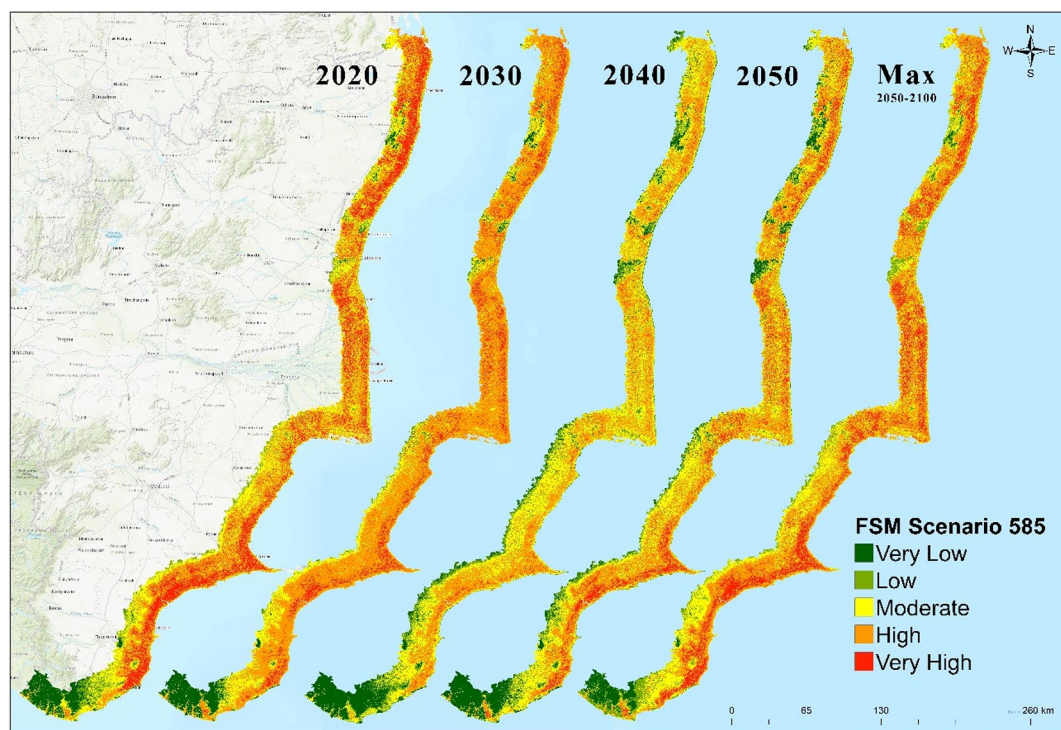


Fig. 21 Flood susceptibility map SSP-585

greater in close coastal areas compared to regions located further away. The Scenario Max(2050–2100) reveals the highest average rainfall recorded throughout the period from 2050 to 2100. This finding suggests that a substantial proportion of the study area—nearly 91%—displays a high level of susceptibility, classified as very high and high vulnerability.

Discussion

Urban growth is being driven by a number of important factors, such as rural–urban migration, employment opportunities, natural development, and urban redefinition (World Bank 2007; Kafy et al. 2021). A study conducted on the coastal region of Tamil Nadu revealed significant changes in land use and land cover (LULC), with an increase in developed areas and a decrease in vegetation cover. The growth of the IT industries and the development of the harbour resulted in the expansion of commercial areas. With an expansion to the west, land has been developed with the construction of residential, commercial, and metro structures and industrial estates. It was discovered that climatic factors, such as precipitation during the northeast monsoon, influenced the region's LULC. The study also highlighted the impact of floods, cyclones, and rising sea levels on the area. Encroachments have led to changes in lake systems like Pulicat, Puzhal, and Chembarambakkam. The

study predicts that Kanchipuram, Villupuram, Cuddalore, and Pondicherry regions will likely witness further changes in LULC. The Cuddalore area is prone to floods and cyclones, resulting in vegetation loss and waterlogging. Numerous high-rise residential and commercial buildings have been built in Pondicherry due to the city's quick growth as a tourist destination, and highway construction, port development, and airport expansion have also been a part of this development. In short, population growth is the most common factor explaining LULC changes in Pondicherry (Nagamani and Ramachandran 2003).

Although the conversion rate of agricultural land to other purposes like industries and building development has grown rapidly over the past several years, the Cuddalore coastline zone still maintains more agricultural land than any additional land use/land cover (Muthusamy et al. 2010). Nagapattinam is a coastal city that has been severely affected by coastal erosion. The result has been a significant loss of coastal land, including urban and agricultural land. This loss of land has led to a shift in land use patterns. Nagapattinam has experienced several natural disasters recently, including floods and cyclones. These events have led to significant changes in land use patterns as people are forced to relocate and rebuild in safer areas. Vedaranyam identified intensive aquaculture and salt pan activities along the coast as two main

contributors to coastal zone instability (Sathiya Bama et al. 2020). In Ramanathapuram, the agricultural land is converted to scrubland, and the invasion of *Prosopis juliflora* resulted in a decrease in water bodies in the study area. The changes in land use pattern were mainly attributed to the growth of scrub land, as no significant alterations were made in built-up areas, industrial zones, and plantations (Sarojini Devi 2013).

According to the findings, Tuticorin is undergoing rapid urbanisation, resulting in the fast depletion of salt pans and fertile lands. Developing onshore and offshore constructions, port facilities, thermal power plants, copper smelters, petrochemical industries, and other activities have altered the land use pattern. Since establishing the port, Tuticorin has become one of India's major industrial centres with a high economic growth rate. All these man-made activities have considerably impacted the land use pattern in the region (Mithra and Bashkaran 2018). The coastal region of Kanyakumari has seen a trend where one-third of fallow lands are being converted into settlements, which reflects the area's significant population growth. The fallow lands close to major coastal settlements are being utilised for infrastructure and construction to support the rapid urban expansion. Nevertheless, due to the conversion of cultivable land and deforestation of plantation and woody vegetation in various parts of the study area, there has been an increase in the agricultural land area. A few fallow lands have also been transformed into salt pans (Kaliraj et al. 2017).

Climate change and LULC changes have worsened the risk of flooding and socioeconomic impacts. Climate change has led to higher rainfall intensity and extreme weather events, which increases the risk of flooding and economic damage (Bukovsky et al. 2021b). Meanwhile, different urbanisation patterns have had varying effects on the economy and ecology. Low-carbon urbanisation patterns have promoted economic development and improved the urban environment. In contrast, high-carbon urbanisation patterns have led to more extensive growth in population and GDP, increasing the exposure to flooding risk. Additionally, the distribution and transformation of land cover have caused varying levels of interaction with flood hazards, leading to significant spatial variations in the resulting socioeconomic risks (Yang et al. 2023). Based on the findings, it is evident that the projections of flooding susceptibility models (FSM) are increasing in all scenarios compared to the IMD. Furthermore, it was observed that the amount of rainfall in 2050 would decrease compared to 2020, 2030, 2040, and the Max(2050–2100). The reason for the decrease of SSP in 2050 would depend on the specific SSP and the factors that influence the scenario's evolution.

Adopting low-carbon urbanisation patterns such as SSP126 and SSP245 has positively impacted urban ecology and, consequently, led to economic growth. Conversely, SSP370 and SSP585, which entail increased carbon emission levels, have resulted in significant population density and GDP growth (Peng and Li 2021). Consequently, the population and GDP exposure to flood risk will increase significantly. Additionally, varying land cover distribution and transformation will result in changing population and GDP interaction with flood hazards, thereby creating significant spatial heterogeneities in the induced socioeconomic risks. As a result of the combination of the aforementioned factors, both the population and GDP in the Future are at greater risk of being affected by flooding, and exposure to floods varies in space and time (Arnell and Lloyd-Hughes 2014; Kim et al. 2019). Additionally, the hydrological and socioeconomic variables, such as rainfall intensity, population density, and GDP per capita, were subject to significant fluctuations during the processes of climate change and urbanisation. As a result, the risks associated with flood exposure were highly uncertain for the Future (Yang et al. 2023).

The trend of the SSP 126 scenario shows an increasing risk of flooding over time, with a higher proportion of areas falling into the moderate to very high-risk categories, while the proportion of areas with lower flood risk is decreasing, but with some fluctuations and variations between time periods. The trend of the SSP 245 scenarios indicates an increasing risk of flooding over time, with a higher proportion of areas falling into the moderate to very high-risk categories and the proportion of areas with lower flood risk decreasing. The risk of flooding is predicted in scenario SSP 370 to increase over time, with a higher proportion of areas falling into the moderate to very high-risk categories, while the proportion of areas with lower flood risk is decreasing. The percentage of areas with a very high risk of flooding is predicted to decrease in the short term but increase in the long term. The SSP 585 scenarios of flood areas in the "very low" and "low" categories are decreasing, while the percentages in the "moderate", "high", and "very high" categories are increasing, indicating a higher risk of flooding over time.

The findings of this study show that the risk of flooding is increasing over time, and the proportion of areas with lower flood risk is decreasing. The severity of the flood risk is also predicted to grow, as the proportion of regions falling into moderate to very high-risk categories increased in all scenarios. These findings also suggest that there are some fluctuations and variations in the trends of different scenarios. For instance, with the SSP 370 scenario, the model predicted a short-term decrease in the percentage of areas with very high risk, followed by a

long-term increase. Such variations may depend on the assumptions and models used in the prediction process.

The study focused on the various factors that affect flood susceptibility and how ML models can effectively evaluate flood susceptibility in different regions with similar environmental aspects and model input parameters while taking into account hydrogeological and topographical differences and considering dynamic parameters such as LULC and rainfall, which can help identify the efficacy of flood exposure mitigation. The proposed method can also be used in areas with limited data to derive flood susceptibility maps. The study's findings can be used to map vulnerability to other flood-related catastrophes, including poor water quality, siltation, shifting shorelines, and loss of vegetation. This research can improve the decision-making process for activities related to flood preparedness.

Conclusion

The study aimed to find a reliable method for detecting flood-prone areas and compared the optimisation of random forest algorithms. The analysis revealed that Bayesian optimisation performed better. The study utilised GEE due to its cloud platform, user-friendly interface, efficient scripting language, adaptability, and flexibility for identifying Land Use and Land Cover (LULC). The random forest classifier yielded promising results, and CA-MC produced results with acceptable accuracies. The study examined LULC for 2000, 2010, and 2020 and forecasted it for 2030, 2040, and 2050. The future climate projections for four scenarios of SSP-126, 245, 370, and 585 used the average annual precipitation of CMIP6 models. The study primarily focussed on the extent, direction, spatial and quantitative consequences of flooding. The major factors in development were transition and rapid industrialisation, and the significant flooding problems were unplanned urbanisation and encroachment, ruined water bodies, and clogged stormwater. The study identified 12 critical parameters for flood susceptibility and recommended including a buffer zone to limit development in these areas. Our results are supported by the distinct emission assumptions embedded in the CMIP6 scenarios. Each scenario (SSP126, 245, 370, 585) assumes different levels of greenhouse gas emissions based on varying socioeconomic and policy conditions. SSP126 anticipates low emissions due to sustainable practices, while SSP585 reflects high emissions in a scenario with limited climate mitigation efforts. The deviation in our projection to 2050 aligns with the emission trajectories prescribed by these scenarios, emphasising the crucial role of socioeconomic factors and policy choices in shaping future climate outcomes. The study, however, has limitations like the need for hydrological modelling, such as

runoff, which could have produced more accurate results on flood occurrence with respect to time. To anticipate changes in the study area, it is necessary to assess the potential impact of climate change and land cover changes on flooding. The findings of this study can be applied to the development of policies for resource management, public planning, and sustainable development by decision-makers, including municipal authorities and project planners. Land use and land cover changes are complex and dynamic processes that have far-reaching effects on the environment, society, and economy. The patterns and trends of these changes, both globally and in India, highlight the need for sustainable land management policies that balance economic development with environmental conservation. Future scenarios of land use and land cover changes underscore the importance of taking action to mitigate human activities impact on the environment. The resulting flood susceptibility maps can be helpful for flood planning and mitigation efforts, helping local authorities to identify vulnerable areas and issue appropriate warnings and evacuation orders.

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Author contributions

D.A. Material preparation, data collection, Original draft preparation were performed, S.S. Conceptualization and design of the study were conducted, P.K.S.S. N.M.R. N.J. data collection data curation and analysing results were conducted, A.A.B. J.M. M.M.A. H.G.A. Visualization, Analyzing Results, Reviewing, Editing the Article, Funding Acquisition, were performed, Writing—review and editing. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author or upon reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

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